

MODEL QUESTION PAPER — II

MATHEMATICS — CLASS IX (CBSE)

Time Allowed: 3 Hours

Maximum Marks: 80

Subject: Mathematics

Class: IX

Chapters Covered: Ch 3 – The World of Numbers | Ch 6 – Measuring Space: Perimeter and Area | Ch 1 – Coordinates (Revision) | Ch 2 – Linear Polynomials (Revision) | Ch 4 – Algebraic Identities (Revision) | Ch 5 – Circles (Revision)

General Instructions:

1. This question paper has **5 sections** — A, B, C, D and E.
2. Section A: 20 MCQs of **1 mark** each.
3. Section B: 5 questions of **2 marks** each.
4. Section C: 6 questions of **3 marks** each.
5. Section D: 4 questions of **5 marks** each.
6. Section E: 3 Case-Based questions of **4 marks** each.
7. All questions are compulsory. Internal choices are marked with **OR**.
8. Use $\pi = 22/7$ unless stated otherwise.
9. Draw neat, labelled diagrams wherever required. Calculator not permitted.

SECTION A — Multiple Choice Questions (20 × 1 = 20 Marks)

Each question carries 1 mark. Choose the most appropriate option.

1. Which of the following is an irrational number?

- | | |
|----------------|----------------|
| (a) 0.333... | (b) $\sqrt{4}$ |
| (c) $\sqrt{5}$ | (d) $22/7$ |

2. The decimal expansion of a rational number is always:

- | | |
|---------------------------------------|----------------------|
| (a) non-terminating and non-repeating | (b) terminating only |
| (c) terminating or repeating | (d) always repeating |

3. Which of the following is NOT a rational number?

- | | |
|----------------|----------------|
| (a) 0 | (b) $-7/2$ |
| (c) $\sqrt{9}$ | (d) $\sqrt{7}$ |

4. $0.\overline{6}$ (i.e., 0.666...) expressed as a fraction is:

- | | |
|-----------|----------------------|
| (a) $2/3$ | (b) $6/9$ |
| (c) $3/5$ | (d) Both (a) and (b) |

5. The product of two irrational numbers is:

- | | |
|-----------------------|----------------------------|
| (a) always irrational | (b) always rational |
| (c) always an integer | (d) rational or irrational |

6. The decimal expansion of $13/250$ is:

- | | |
|-----------------------------------|-------------------------------|
| (a) terminating | (b) non-terminating repeating |
| (c) non-terminating non-repeating | (d) cannot be determined |

7. Between two consecutive rational numbers, the number of rational numbers that exist is:

- | | |
|--------------|---------------|
| (a) 0 | (b) finite |
| (c) infinite | (d) exactly 1 |

8. The proof technique used to show that $\sqrt{2}$ is irrational is called:

- | | |
|----------------------------|----------------------------|
| (a) Mathematical induction | (b) Proof by contradiction |
|----------------------------|----------------------------|

(c) Direct proof

(d) Proof by exhaustion

9. The circumference of a circle with radius 7 cm is: (Use $\pi = 22/7$)

(a) 22 cm

(b) 44 cm

(c) 154 cm

(d) 88 cm

10. The area of a circle with diameter 14 cm is: (Use $\pi = 22/7$)

(a) 44 cm²

(b) 88 cm²

(c) 154 cm²

(d) 616 cm²

11. The arc length of a sector with radius 6 cm and central angle 60° is:

(a) 2π cm

(b) π cm

(c) 4π cm

(d) 6π cm

12. The area of a sector with radius 7 cm and angle 90° is: (Use $\pi = 22/7$)

(a) 38.5 cm²

(b) 77 cm²

(c) 19.25 cm²

(d) 154 cm²

13. The semi-perimeter of a triangle with sides 5 cm, 6 cm and 7 cm is:

(a) 6 cm

(b) 9 cm

(c) 18 cm

(d) 36 cm

14. A parallelogram and a rectangle on the same base and between the same parallels have:

(a) equal perimeters

(b) equal areas

(c) equal diagonals

(d) equal heights only

15. A median of a triangle divides it into two triangles of:

(a) equal perimeter

(b) equal angles

(c) equal area

(d) equal base

16. The slope of the line $y = -3x + 5$ is:

(a) 5

(b) 3

(c) -3

(d) -5

17. The distance between points (0, 0) and (-3, 4) is:

(a) 3 units

(b) 4 units

(c) 5 units

(d) 7 units

18. $(a + b)^3 - (a - b)^3$ simplifies to:

(a) $2a^3$

(b) $2b^3$

(c) $2a^3 + 6ab^2$

(d) $6a^2b + 2b^3$

19. The angle subtended by an arc at the centre is 100°. The angle it subtends at any point on the major arc is:

(a) 100°

(b) 200°

(c) 50°

(d) 80°

20. A chord of a circle of radius 13 cm is at a distance of 5 cm from the centre. The length of the chord is:

(a) 12 cm

(b) 24 cm

(c) 10 cm

(d) 26 cm

SECTION B — Short Answer Questions (5 × 2 = 10 Marks)

Each question carries 2 marks.

21. Convert $0.2\overline{35}$ (i.e., 0.23535...) into the form p/q , where p and q are integers and $q \neq 0$.

22. Without performing long division, state whether the decimal expansion of $7/20$ is terminating or non-terminating repeating. Give a reason. If terminating, find the decimal.
23. The perimeter of a semicircle is 36 cm. Find its radius. (Use $\pi = 22/7$)
24. The sides of a triangle are 13 cm, 14 cm and 15 cm. Find its semi-perimeter and hence use Heron's formula to find its area.
25. Show that $\sqrt{3}$ is an irrational number using the method of proof by contradiction.

SECTION C — Short Answer Questions ($6 \times 3 = 18$ Marks)

Each question carries 3 marks. Internal choice is given where indicated.

26. The sides of a triangular plot are in the ratio 3 : 5 : 7 and its perimeter is 300 m.

- (i) Find the lengths of the three sides.
 - (ii) Find the semi-perimeter.
 - (iii) Using Heron's formula, find the area of the triangular plot.
- (Use $\sqrt{3} \approx 1.73$)

27. A sector has a radius of 14 cm.

- (i) Find the length of arc AB if the central angle $\angle AOB = 90^\circ$.
- (ii) Find the area of the sector.
- (iii) Find the perimeter of the sector (arc + two radii). (Use $\pi = 22/7$)

28. State the C/D ratio for circles and define π . Why is π irrational?

An athletics track has two straight sections of 84.39 m each and two semicircular ends with innermost radius 36.5 m. Each lane is 1.22 m wide.

- (i) What distance does an athlete in Lane 1 (running 0.3 m from inner border) cover in one lap?
- (ii) Calculate the stagger for the runner in Lane 2 compared to Lane 1.

OR

A chord of a circle of radius 10 cm subtends a 90° angle at the centre.

- (i) Find the area of the minor sector (angle 90°). (Use $\pi = 3.14$)
- (ii) Find the area of the triangle formed by the two radii and the chord.
- (iii) Hence find the area of the minor segment.

29. A spice trader starts the week with a fortune of ₹1200. By Tuesday he incurs a debt of ₹850. On Friday he makes a profit of ₹600.

- (i) Express his financial journey as a sequence of integer additions.
- (ii) What is his final financial standing?
- (iii) Write a linear polynomial $f(d)$ for the total money if he earns ₹150 every day starting from his Tuesday balance. For which value of d does $f(d) = 0$?

30. Prove that a median of a triangle divides it into two triangles of equal area. Draw a neat figure and write the complete proof.

- (i) In $\triangle ABC$, D is the midpoint of BC and P is any point on AD.
- (ii) Show that $\text{area}(\triangle ABP) = \text{area}(\triangle ACP)$. Use the theorem you just proved.

OR

O is any point on diagonal PR of parallelogram PQRS. Prove that:

$$\text{area}(\triangle PSO) = \text{area}(\triangle PQO).$$

31. The midpoints of the sides of a quadrilateral are joined in order to form another quadrilateral. Using the property of medians and area of triangles, prove that the area of the inner quadrilateral is half the area of the outer quadrilateral.

(Hint: Draw the diagonals of the outer quadrilateral and use the theorem on medians.)

SECTION D — Long Answer Questions (4 × 5 = 20 Marks)

Each question carries 5 marks. Internal choice is given where indicated.

32. (i) Prove that $\sqrt{2}$ is irrational using the method of proof by contradiction.
(ii) On a number line, construct a line segment of length $\sqrt{5}$ using the Baudhāyana–Pythagoras theorem. Describe the steps clearly.
(iii) Find 5 rational numbers between $\frac{3}{5}$ and $\frac{4}{5}$.
(iv) State whether 1.010010001000010000... (each time one more zero is added before the digit 1) is rational or irrational. Justify your answer.

OR

- (i) Convert each of the following into p/q form: (a) 2.1 (b) 0.67 (c) 3.142
(ii) Without performing division, identify which of the following decimals are terminating: (a) $\frac{17}{8}$ (b) $\frac{15}{12}$ (c) $\frac{64}{455}$ (d) $\frac{29}{343}$
(iii) The number 142857 is a cyclic number (it is the repeating block of $\frac{1}{7}$). Multiply it by 1, 2, 3, 4, 5, 6 and explain the pattern you observe.
(iv) Prove that 0.9 (i.e., 0.999...) is exactly equal to 1.
33. An isosceles triangle has equal sides of length 10 cm each and a base of 12 cm.
(i) Find the area using the formula $\frac{1}{2} \times \text{base} \times \text{height}$.
(ii) Verify your answer using Heron's formula.
(iii) A square garden has the same area as this triangle. Find the side of the square.
(iv) Compare the perimeters of the triangle and the square. Which has the larger perimeter?
(v) Now, if the square is transformed into a rectangle with the same area and one side of 4 cm, find the other side and its perimeter.

OR

- In a parallelogram ABCD, P and Q are any two points on side AB.
(i) Show that $\text{area}(\triangle PCD) = \text{area}(\triangle QCD)$.
(ii) Given that the parallelogram has base 12 cm and height 8 cm, find its area.
(iii) A triangle DEF is drawn between the same parallels as ABCD with $DE = AB = 12$ cm. Find $\text{area}(\triangle DEF)$ and compare it with $\text{area}(ABCD)$.
(iv) Prove that the diagonals of a parallelogram divide it into four triangles of equal area.
34. A wheel of a bicycle has a diameter of 56 cm.
(i) Find the circumference of the wheel. (Use $\pi = \frac{22}{7}$)
(ii) How far does the bicycle travel in 500 complete revolutions of the wheel?
(iii) A farmer has a circular field of radius 14 m. He wants to fence it and also sow seeds. Fencing costs ₹25 per metre and seeds cost ₹50 per square metre. Find the total cost.
(iv) A sector of the same circular field (with radius 14 m) is reserved for a pond with central angle 90° . Find the area of the pond and the remaining area available for cultivation.

35. Answer the following problems on area and perimeter:

- (i) Two circles of equal radius r are placed so that each passes through the centre of the other. Find the perimeter of the region enclosed by the two circles in terms of r . (Do NOT include the region inside either circle.)
(ii) In Fig., points P and Q are connected by two paths: Path a is a large semicircle and Path b + c + d consists of three smaller semicircles. Show algebraically that the two paths have equal length.
(iii) Nine identical small rectangles are arranged to fill a larger rectangle of area 72 cm^2 . Find the perimeter of each small rectangle.

(iv) A quadrilateral has sides 9 cm, 40 cm, 28 cm and 15 cm. It has a right angle between the sides of 9 cm and 40 cm. Find its area. (Hint: Split into a triangle and another triangle using a diagonal.)

OR

(i) Three problems on fitting congruent shapes: Rectangle ABCD has sides a and b . Rectangle PQRS has sides $2a$ and $2b$. Show that PQRS has 4 times the area of ABCD.

(ii) A chord BC of a larger circle touches a smaller concentric circle at A. If $BC = l$, show that the area of the green ring between the two circles is $\pi l^2/4$.

(iii) Semicircles are drawn outward on all three sides of a right-angled triangle with legs 3 cm and 4 cm. Show that the sum of the areas of the two smaller semicircles equals the area of the largest semicircle (a generalisation of the Baudhāyana–Pythagoras theorem).

(iv) Using Brahmagupta's formula, find the area of a cyclic quadrilateral whose sides are 5 cm, 6 cm, 7 cm and 8 cm.

SECTION E — Case-Based Questions (3 × 4 = 12 Marks)

Each case study is followed by 4 sub-questions of 1 mark each. Attempt all sub-questions.

Case Study 1 — The Ishango Bone and Number Systems (Questions 36–39)

The Ishango bone (c. 20,000 BCE) found near the Congo River contains tally marks in groups 11, 13, 17, 19 — the prime numbers between 10 and 20. This is one of the earliest recorded examples of mathematical thought. Brahmagupta (628 CE) later formalised zero, negative numbers and rational arithmetic. He defined a rational number as any number expressible as p/q ($q \neq 0$), and gave rules for their arithmetic. His work laid the foundation for the modern real number system: $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$, where $\mathbb{Q} = \mathbb{Z} \cup \mathbb{Q}^+$ and $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$.

36. Write the next three prime numbers after 19 (i.e., the primes that would appear next on a hypothetical extended Ishango bone). (1 mark)

37. Brahmagupta stated: 'the product of two debts is a fortune.' Express this as an algebraic rule. Evaluate: $(-8) \times (-7)$ using this rule. (1 mark)

38. Using Brahmagupta's rules for rational numbers, simplify: $(-3/4) \times (8/9) \div (-2/3)$. (1 mark)

39. Identify which set ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{I}$) each of the following belongs to: (a) -5 (b) $3/7$ (c) $\sqrt{11}$ (d) $0.252525\dots$ (1 mark)

Case Study 2 — The 400 m Athletics Track (Questions 40–43)

An athletics track has two straight sections of 84.39 m each and two semicircular curves at each end. The innermost semicircle has radius 36.5 m. Each lane is 1.22 m wide. Athletes run at 0.3 m from their lane's inner edge. The track is designed so that each athlete runs exactly 400 m in Lane 1. Runners in outer lanes must start ahead of those in inner lanes by an amount called the 'stagger', to compensate for the greater circumference of the outer semicircles. The stagger between adjacent lanes is the same because the lane width is uniform. (Use $\pi = 22/7$)

40. A runner in Lane 1 runs at an effective radius of $36.5 + 0.3 = 36.8$ m from the centre. Verify that her total distance in one lap is 400 m (straight sections + two semicircles). (1 mark)

41. A runner in Lane 2 runs at radius $36.5 + 1.22 + 0.3 = 38.02$ m. Find the total distance she would cover in one lap without a stagger. (1 mark)

42. Find the stagger needed for the Lane 2 runner so that both athletes cover exactly 400 m. (1 mark)

43. If all lanes have the same width (1.22 m), is the stagger between consecutive lanes always equal? Calculate the stagger between Lane 2 and Lane 3 and compare it with the stagger between Lane 1 and Lane 2. (1 mark)

Case Study 3 — A Farmer's Circular Field (Questions 44–47)

A farmer owns a circular field of radius 21 m. He divides it as follows: A triangular portion with vertices at the centre O and two points A, B on the circumference ($\angle AOB = 60^\circ$) is reserved for a water tank. A second sector with

$\angle COD = 120^\circ$ (radius $OC = OD = 21$ m) is used for a flower garden. The remaining area of the circle is used for growing wheat. The farmer also wants to put a fence around the entire circular field and along the two straight edges (OA and OB) of the triangular water-tank region. (Use $\pi = 22/7$ and $\sqrt{3} \approx 1.73$)

44. Find the total area of the circular field. Also find the area of the sector reserved for the water tank (angle 60°). (1 mark)

45. Find the area of the triangular region OAB (with $\angle AOB = 60^\circ$ and $OA = OB = 21$ m). Hence find the area of the minor segment cut off by chord AB. (1 mark)

46. Find the area of the flower-garden sector (angle 120°) and the area left for wheat cultivation. (1 mark)

47. Find the total length of fencing required: the circumference of the full circle plus the two straight edges OA and OB of the water-tank triangle. (1 mark)

Section	Type	No. of Questions	Marks per Q	Total
A	MCQ	20	1	20
B	Short Answer I	5	2	10
C	Short Answer II	6	3	18
D	Long Answer	4	5	20
E	Case-Based	3 cases $\times$ 4 sub-Qs	1	12

Grand Total

80

■ All the Best! ■

This model paper covers: The World of Numbers • Measuring Space: Perimeter and Area | Plus revision of Coordinates, Linear Polynomials, Algebraic Identities & Circles | Based on CBSE Class IX Syllabus — Ganita Manjari Part I